Bike count prediction as a time-series supervised learning task

We compare required model families for +1h and +24h, while keeping lag features leakage-safe.

1 Direct targets

For horizon h, target = BikeCount.shift(-h). The +24h task predicts one hour 24 hours ahead, not a 24h sum.

2 Feature engineering

Target-hour calendar and weather, cyclical hour/month encodings, robust weather buckets, lag and rolling count features.

3 Temporal validation

Last 61 days as holdout. We avoid random splits because they leak time information through lag features.

Models tried

- Linear: Ridge regression
- Tree-based: RandomForest and XGBoost variants
- Neural network: MLP with scaled inputs/target

8,759
clean rows

26
+1h features

61d
holdout

Submitted model/notebook evaluate +1h only; slides compare both requested horizons.

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Boosted trees were strongest; XGBoost tuned log1p is our +1h submission model

Validation MSE on the last 61 days; lower is better. Hidden-test MSE can be added after Wednesday's run.

Model	+1h MSE	+24h MSE
naive horizon lag	22,138	60,513
naive lag168	45,525	45,525
Ridge	11,582	28,223
RandomForest	4,035	25,438
XGBoost	3,455	24,373
XGBoost log1p	3,320	22,595
XGBoost tuned	3,332	22,660
XGBoost tuned log1p	**3,220**	**19,988**
MLP	9,303	39,635

Selected model

XGBoost tuned log1p

+1h validation MSE: 3,220

Interpretation

XGBoost captures calendar, weather and lag interactions better than Ridge or MLP here.

Wednesday update

Run notebook_group18.ipynb and report the final-cell value. Hidden-test MSE: TBD.